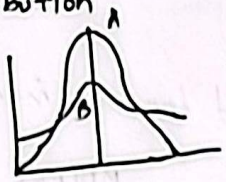
Measures of Dispersion

Measures of dispersion gives the degree of scatterness about the central location. The main purpose is it gives basis of comparison between two or more frequency distribution



→ Here the distribution of A and B have the same mean $\bar{m}$, but they differ in variation. Such variation is called dispersion.

⊛ Characteristic of an ideal measure of dispersion same as measure of location.

classification:

A) Absolute measure: express as same unit as the original data.
   i) Range
   ii) Quantile deviation
   iii) Standard deviation
   iv) mean deviation

B) Relative measure: pure numbers (ratio, percentage), used when comparing datasets with different unit/scale or wide number differ.
   i) Co-efficient of quantile deviation
   ii) Co-efficient of variation
   iii) Co-efficient of mean deviation

□ Range: The range of a set of values is the difference between the highest and lowest value in the set. Used in measurement of share market fluctuation.

□ Quantile deviation: Quantile divide the observation into 4 equal parts when the observation are arranged in order of magnitude. If the Q_1 and Q_3 denotes the first and third quantile, then

$$\text{Quantile deviation} = \frac{Q_3 - Q_1}{2}$$

| It also called semi-interquantile range

□ Standard deviation: The arithmetic mean of the squares of the deviation of the given observation of the given observations from their arithmetic mean, is known as variance. The positive square root of the variance is standard deviation.

If $x_1, x_2, x_n \ldots$ be n observations of a variable the standard deviation is defined by,

$$\sigma_x = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

In case of frequency distribution,

$$\sigma_x = \sqrt{\frac{1}{n}\sum_{i=1}^{k} f_i (x_i - \bar{x})^2} \quad ; \text{ where } n = \sum_{i=1}^{k} f_i$$

working formula,

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum(x_i^2 - 2x_i\cdot\bar{x} + \bar{x}^2)$$
$$= \sum x_i^2 - 2\bar{x}\sum x_i + \sum \bar{x}^2$$
$$= \sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2$$
$$= \sum x_i^2 - n\bar{x}^2 = \sum x_i^2 - n\left(\frac{\sum x_i}{n}\right)^2$$
$$= \sum x_i^2 - \frac{\sum x_i^2}{n}$$

Then,

$$\sigma_x^2 = \frac{1}{n}\cdot\left(\sum x_i^2 - \frac{\sum x_i^2}{n}\right)$$
$$= \frac{\sum x_i^2}{n} - \bar{x}^2$$

1

Mean deviation :-

Mean deviation of a set of quantities $m_1, m_2 \dots m_n$ is defined by,

mean deviation from mean, $M.D_{(\bar{m})} = \frac{1}{n}\sum_{i=1}^{n} |m_i - \bar{m}|$ / $MD(m_e) = \frac{1}{n}\sum_{i=1}^{n} |m_i - m_e|$ / $MD(m_0) = \frac{1}{n}\sum_{i=1}^{n} |m_i - m_0|$

(mean deviation from median)

Relative measure

i) Co-efficient of Quantile Deviation : If Q_1 and Q_3 are the first and third quantile.

$$C \text{ of } QD = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

ii) Co-efficient of Variation : $C.V = \frac{S_m}{\bar{m}} \times 100$ | $S_m \to$ standard deviation | $\bar{m} \to$ mean

iii) Co-efficient of mean deviation :

a) $CMD(\bar{m}) = \frac{MD(\bar{m})}{\bar{m}} \times 100$

b) $CMD(m_e) = \frac{MD(m_e)}{m_e} \times 100$

c) $CMD(m_0) = \frac{MD(m_0)}{m_0} \times 100$

Why SD is the most important measure of dispersion?

merit:
- Popular measure
- Further algebraic
- Works with all the observation
- Identify outliers or extreme values.
- less affected by sample fluctuations.

Let $\bar{m} = 300$ and $SD = 15$

$\bar{m} \pm 3SD$ < $300+45 = 345$, $300-45 = 255$

Range : 255 - 345

If the other observation are outside of this this will be the outlier

- ✓ $\bar{m} \pm SD \to 68.26\%$ observation will be in this range
- ✓ $\bar{m} \pm 2SD \to 95.44\%$ observation will be in this range
- ✓ $\bar{m} \pm 3SD \to 99.74\%$ observation will be in this range

If the whole observation in this range then this is normally distributed data.

Z score :-

$$Z = \frac{m - \bar{m}}{SD}$$

more z value is better

stat → $\bar{m} = 75$, $SD = 10$, $m = 82$

agri → $\bar{m} = 80$, $SD = 14$, $m = 86$

$Z_{stat} = \frac{82-75}{10} = 0.7$ and $Z_{agri} = \frac{86-80}{14} = 0.43$

Theorem 1 : Standard deviation is the least possible root mean square deviation.

Let $m_1, m_2, \dots m_k$ are the values of k observations with corresponding frequency $f_1, f_2, \dots f_k$, also, let $\bar{m}$ be the mean of the observation and A is any arbitary numbers.

We know, $S'^2 = \frac{1}{n}\sum_{i=1}^{k} f_i (m_i - A)^2$

$= \frac{1}{n}\sum_{i=1}^{k} f_i \{(m_i - \bar{m}) + (\bar{m} - A)\}^2$

$= \frac{1}{n}\sum_{i=1}^{k} f_i (m_i - \bar{m})^2 + 2(\bar{m} - A)\sum_{i=1}^{k} f_i (m_i - \bar{m}) + n(\bar{m} - A)^2$

$= \frac{1}{n}\sum_{i=1}^{k} f_i (m_i - \bar{m})^2 + 0 + n(\bar{m} - A)^2 \quad \left| \sum_{i=1}^{k} (m_i - \bar{m}) = 0 \right.$

$= S^2 + n(\bar{m} - A)^2$

So, $S^2 < S'^2$ (as $n(\bar{m} - A)^2$ always positive)

Theorem 2: The standard deviation of the first natural number is $\sqrt{\frac{(n^2-1)}{12}}$

Proof: The first n natural numbers are $1, 2, 3 \ldots n$.

• mean of the first n natural numbers $\bar{x} = \frac{1+2 \ldots +n}{n} = \frac{n(n-1)}{2} \cdot \frac{1}{n} = \frac{n-1}{2}$

• and, $\sum x_i^2 = 1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$

We know, $s_1^2 = \frac{1}{n}\sum (x_i - \bar{x})^2 = \frac{1}{n}\left(\sum x_i^2 + \frac{\sum x_i^2}{n}\right) = \frac{\sum x_i^2}{n} + \bar{x}^2$

(working formula)

$= \frac{n(n+1)(2n+1)}{6 \cdot n} + \left(\frac{n-1}{2}\right)^2$

$= \frac{n+1}{2}\left(\frac{2n+1}{3} - \frac{n-1}{2}\right)$

$= \frac{n+1}{2} \cdot \frac{n-1}{6} = \frac{n^2-1}{12}$

$\therefore S = \sqrt{\frac{(n^2-1)}{12}}$

Theorem 3: For two observation standard deviation is the half of the range:

Proof: let x_1 and x_2 are two observation ; mean $= \frac{x_1+x_2}{2}$

we know, $s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

$s^2 = \frac{1}{2}\left[\left(x_1 - \frac{x_1+x_2}{2}\right)^2 + \left(x_2 - \frac{x_1+x_2}{2}\right)^2\right]$

$= \frac{1}{2}\left[\left(\frac{x_1-x_2}{2}\right)^2 + \left(\frac{x_2-x_1}{2}\right)^2\right]$

$= \frac{1}{2} \cdot 2\left(\frac{x_1-x_2}{2}\right)^2 \quad \therefore S = \left|\frac{x_1-x_2}{2}\right|$

Theorem 4: If $\bar{x}$ and S be A.M and sd respectively of n non-negative observation, Then $\bar{x}\sqrt{n-1} \geq S$

Moment

A tool to understand nature/form of a distribution. Two types: raw moments and central moment.

If $x_1, x_2, x_3 .. x_n$ be n observations of a variate then the r-th

- raw moment, $\mu_r' = \frac{1}{n}\sum_{i=1}^{n}(x_i - A)^r$, where A is arbitary value.
- central moment, $\mu_r = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^r$, where $\bar{x}$ is A.M.

relationship between raw and central moment,

$$\mu_k = \binom{k}{0}\mu_k' - \binom{k}{1}\mu_{k-1}'(\mu_1') + \binom{k}{2}\mu_{k-2}'(\mu_1')^2 + \cdots\cdots$$

- Deriving first four moments and relationship among them

A) i) $\mu_1' = \frac{\sum(x_i - A)}{n} = \frac{\sum x_i}{n} - \frac{nA}{n} = \bar{x} - A$

From ii) $\mu_1 = \frac{\sum(x_i - \bar{x})}{n} = \frac{\sum x_i}{n} - \frac{n\bar{x}}{n} = \bar{x} - \bar{x} = 0$ (always zero)

B) i) $\mu_2 = \frac{\sum(x_i - \bar{x})^2}{n} =$ **Variance**

$$= \frac{\sum\{(x_i - A) - (\bar{x} - A)\}^2}{n} = \frac{\sum(x_i - A)^2}{n} - 2\cdot\frac{\sum(x_i - A)}{n}\cdot(\bar{x} - A) + \frac{\sum(\bar{x} - A)^2}{n}$$

$$= \mu_2' - 2\mu_1'\mu_1' + \mu_1'^2 = \mu_2' - \mu_1'^2 \qquad \frac{n(\bar{x} - A)^2}{n}$$

From

C) $$\mu_3 = \frac{\sum(x_i - \bar{x})^3}{n} = \frac{\sum\{(x_i - A) - (\bar{x} - A)\}^3}{n}$$

$$= \frac{\sum(x_i - A)^3}{n} - 3\frac{\sum(x_i - A)^2}{n}\cdot(\bar{x} - A) + 3\frac{\sum x_i - A}{n}\cdot(\bar{x} - A)^2 \quad \frac{\sum(\bar{x} - A)^3}{n}\cdot$$

$$= \mu_3' - 3\mu_2'\mu_1' + 3\mu_1'\mu_1'^2 \quad \mu_0'^3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1^3$$

D) $$\mu_4 = \frac{\sum(x_i - \bar{x})^4}{n} = \frac{\sum\{(x_i - A) - (\bar{x} - A)\}^4}{n} = \frac{\sum(x_i - A)^4}{n} - 4\frac{\sum(x_i - A)^3}{n}(\bar{x} - A) + 6\frac{\sum(x_i - A)^2}{n}(\bar{x} - A)^2 - 4\frac{\sum(x_i - A)}{n}(\bar{x} - A)^3 + (\bar{x} - A)^4$$

$[a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4]$

$$= \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

- **Sheppard's correction for moments:**

while calculating moment, it is assumed that all the values of the variable in a class interval lie at the center ie (mid value). But it is distributed. This is known as grouping error, Sheppard suggested some correction:

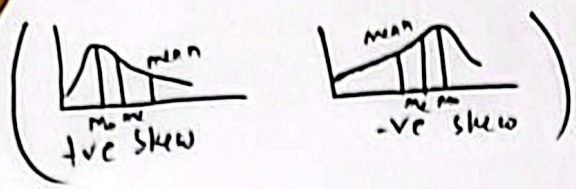
(μ_1 and μ_3 need no correction)

$$\mu_2 \text{ (adjusted)} = \mu_2 - \frac{h^2}{12}$$

$$\mu_4 \text{ (adjusted)} = \mu_4 - \frac{h^2}{2}\mu_2 + \frac{7}{240}h^4$$

• Shape characteristics of a frequency distribution:

a) Skewness: It measure the "degree of asymmetry"



- mean, median, mode are different value
- Q_1 and Q_3 are not equidistance from median.

Skewness $\beta_1 = \frac{\gamma_3^2}{\gamma_2^3}$ [sign is the same as γ_3: γ_3(-ve) → β_1(-ve), γ_3(+ve) → β_1(+ve)]

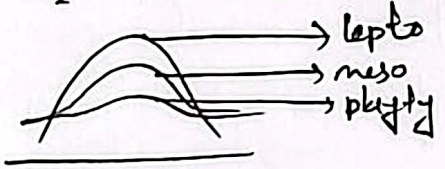
- ✓ $\beta_1 < 0$ → negatively skewed
- ✓ $\beta_1 = 0$ → Normal, symmetric
- ✓ $\beta_1 > 0$ → positively skewed

Karl pearson, $\beta_1 = \frac{\text{Mean} - \text{mode}}{\text{standard deviation}}$

b) Kurtosis: The degree of peakness and flatness of a distribution relative to a normal distribution.

$\beta_2 = \frac{\gamma_4}{\gamma_2^2}$ never negative! Excess of kurtosis $\gamma_2 = \beta_2 - 3$

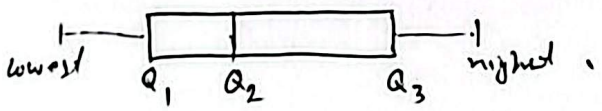
- ✓ For normal distribution, $\beta_2 = 3$ / $\gamma_2 = 0$, the curve called mesokurtic
- ✓ when, $\beta_2 > 3$ / $\gamma_2 > 0$, " " " leptokurtic
- ✓ when, $\beta_2 < 3$ / $\gamma_2 < 0$, " " " platykurtic



⊛ Significance of measuring variation: (purpose)

- To determine the reliability of an average
- To serve as a basis for the control of the variability.
- To compare two or more series
- To facilitate the use of other statistical measure

⊛ Box plot/ whisker plot: gives distribution, location, spread, shape, tail, length and outliers.



CamScanner